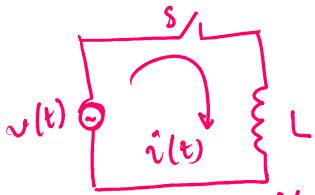
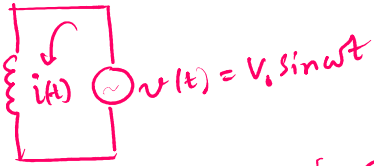


Inductor

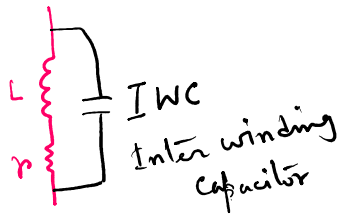


$$v(t) = L \frac{di(t)}{dt}$$

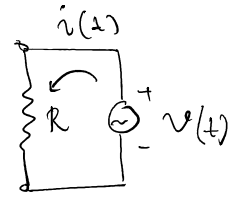
$$\int_0^t i(t) dt = \int_0^t \frac{V_0}{L} \sin \omega t dt$$

$$i(t) = -\frac{V_0}{\omega L} \cos \omega t = \frac{V_0}{\omega L} \sin(\omega t - \pi/2)$$

$$I_0 = \frac{V_0}{\omega L}$$



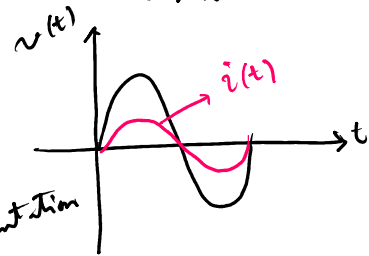
Resistor



$$\begin{aligned} I &\propto V \\ I &= \sigma V \\ IR &= V \end{aligned}$$

$$i(t) \times R = v(t)$$

$$i(t) R = v(t)$$

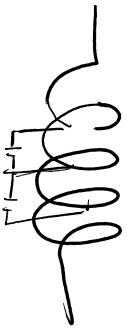


$$v(t) = V_0 \sin \omega t$$

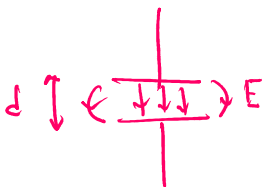
$$i(t) = \frac{V_0}{R} \sin \omega t$$

$$i(t) = I_0 \sin \omega t$$

* Voltage and current are in-phase



Capacitor



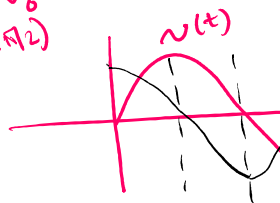
$$J_d(t) \propto \frac{\partial E}{\partial t}$$

$$i_d(t) \propto \epsilon_r \frac{\partial E}{\partial t} = \epsilon_r \frac{\partial}{\partial t} \left(\frac{V_0}{d} \sin \omega t \right) = \frac{A \epsilon_r V_0}{d} \omega \cos \omega t$$

displacement current

$i \downarrow$ $i_{cond} \downarrow$

$$\begin{aligned} i(t) &= \frac{dQ}{dt} = \frac{d}{dt} (C V_0 \sin \omega t) \\ i(t) &= C V_0 \omega \cos \omega t \\ i(t) &= C V_0 \omega \sin(\omega t + \pi/2) \end{aligned}$$



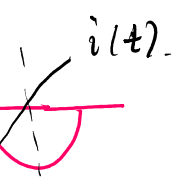
$$= C V_0 \omega \cos \omega t$$

$n\omega t$

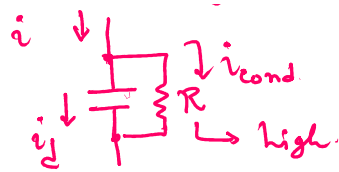
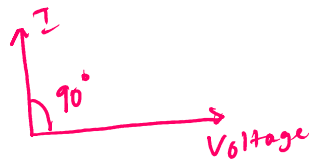
$\sin n\omega t$

$\sin n\omega t$

$n\omega t$



$I_d(t) \approx 0$
current density



$$= C V_G \omega \cos \omega t$$

$$95n4v6e2$$

